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Ansys Fluent Simulation Report

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System Information

Application	Fluent
Settings	3d, double precision, pressure-based, SST k-omega
Version	26.1.0-10262
Source Revision	b6f454dde8
Build Time	Feb 2 2026 16:34:25 EST
CPU	AMD Ryzen 7 6800H with Radeon Graphics
GPU	NVIDIA GeForce RTX 3050 Laptop GPU
OS	Windows

Geometry and Mesh

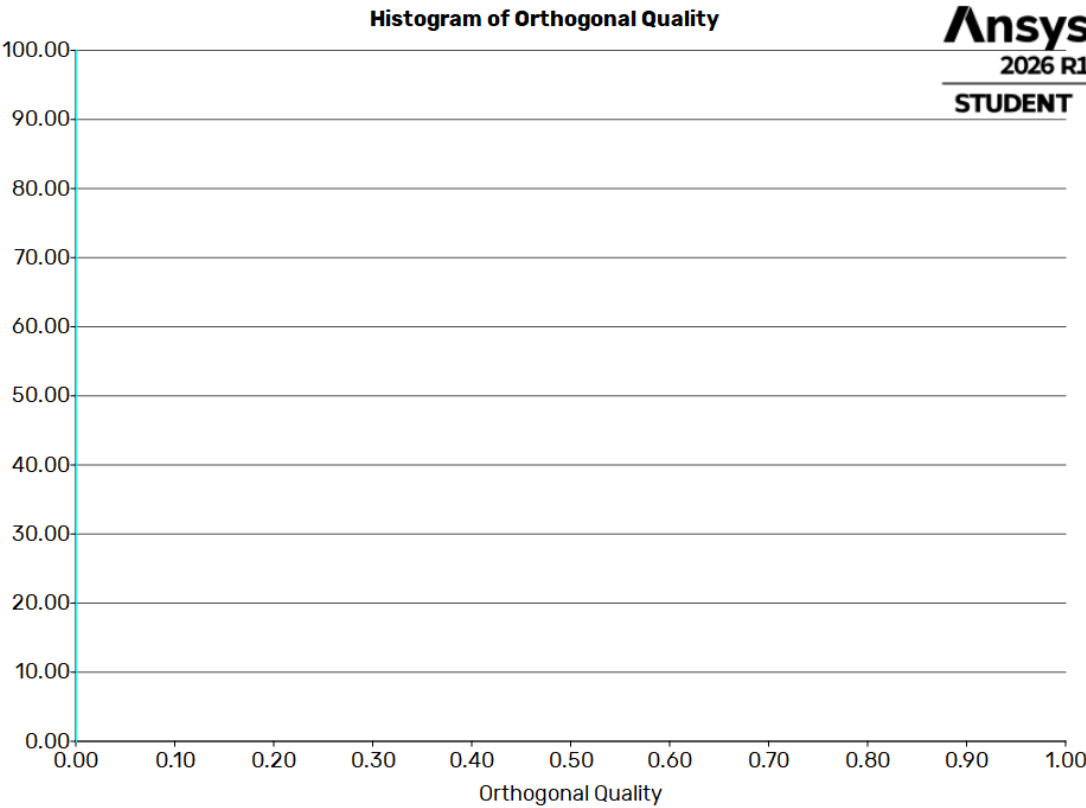
Mesh Size

Cells	Faces	Nodes
107676	241529	33214

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
fluid	Mixed Cell	0.22227342	14.697877
solid11	Tet Cell	0.12993988	20.319746

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	SST k-omega turbulence model

Material Properties

– Fluid	
– water-liquid	
Density	998.2 kg/m^3
Viscosity	0.001003 kg/(m s)
– air	
Density	1.225 kg/m^3
Viscosity	1.7894e-05 kg/(m s)
– Solid	
– steel	
Density	8030 kg/m^3
– aluminum	
Density	2719 kg/m^3

Cell Zone Conditions

– Fluid	
– fluid	
Material Name	water liquid
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Laminar zone?	no
Porous zone?	no
3D Fan Zone?	no

— Solid	
— solid11	
Material Name	steel
Frame Motion?	no
Solid Motion?	no

Boundary Conditions

— Inlet	
— inlet	
Velocity Specification Method	Magnitude, Normal to Boundary
Reference Frame	Absolute
Velocity Magnitude [m/s]	10
Supersonic/Initial Gauge Pressure [Pa]	0
Turbulence Specification Method	Intensity and Viscosity Ratio
Turbulent Intensity [%]	5
Turbulent Viscosity Ratio	10
— Outlet	
— outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	0
Pressure Profile Multiplier	1
Backflow Direction Specification Method	Normal to Boundary
Turbulence Specification Method	Intensity and Viscosity Ratio
Backflow Turbulent Intensity [%]	5
Backflow Turbulent Viscosity Ratio	10
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Radial Equilibrium Pressure Distribution (Unsupported)	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
— Wall	
— fluid-solid11	
Wall Motion	Stationary Wall
Shear Boundary Condition	No Slip
Wall Surface Roughness	Standard
Wall Roughness Height [mm]	0

Wall Roughness Constant	0.5
surface:1	wall
solid11:655	wall
fluid-solid11-shadow	wall

Reference Values

Area	1 m^2
Density	998.2 kg/m^3
Enthalpy	0 J/kg
Length	1 mm
Pressure	0 Pa
Temperature	288.16 K
Velocity	10 m/s
Viscosity	0.001003 kg/(m s)
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	fluid

Solver Settings

– Equations	
Flow	True
Turbulence	True
– Numerics	
Absolute Velocity Formulation	True
– Under-Relaxation Factors	
Pressure	0.3
Density	1
Body Forces	1
Momentum	0.7
Turbulent Kinetic Energy	0.8
Specific Dissipation Rate	0.8
Turbulent Viscosity	1
– Pressure-Velocity Coupling	
Type	SIMPLE
– Discretization Scheme	
Pressure	Second Order
Momentum	Second Order Upwind

Turbulent Kinetic Energy	Second Order Upwind
Specific Dissipation Rate	Second Order Upwind
— Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000
Minimum Turb. Kinetic Energy [m^2/s^2]	1e-14
Minimum Spec. Dissipation Rate [s^-1]	1e-20
Maximum Turb. Viscosity Ratio	100000

Run Information

Number of Machines	1
Number of Cores	4
Case Read	22.757 seconds
Data Read	0.387 seconds
Virtual Current Memory	8.4766 GB
Virtual Peak Memory	8.4956 GB
Memory Per M Cell	72.5015

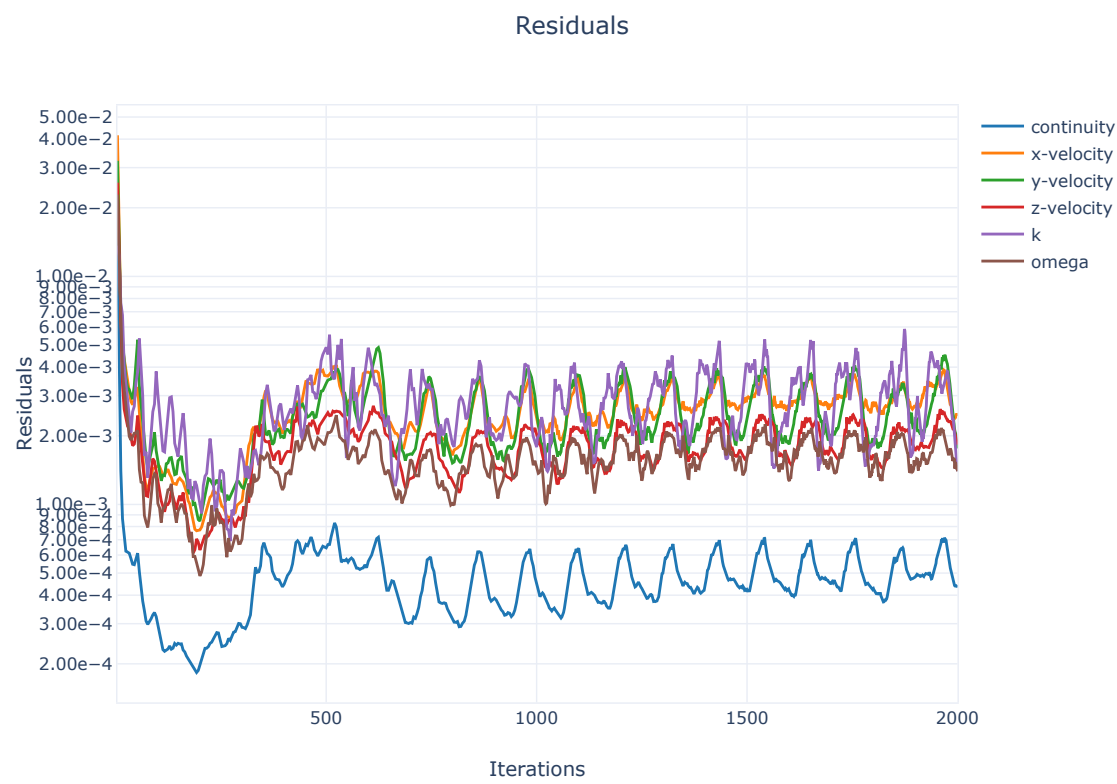
Solution Status

Iterations: 2000

	Value	Absolute Criteria	Convergence Status
continuity	0.0004435178	0.001	Converged
x-velocity	0.00247927	0.001	Not Converged
y-velocity	0.001759697	0.001	Not Converged
z-velocity	0.001828777	0.001	Not Converged
k	0.001399828	0.001	Not Converged
omega	0.001396939	0.001	Not Converged

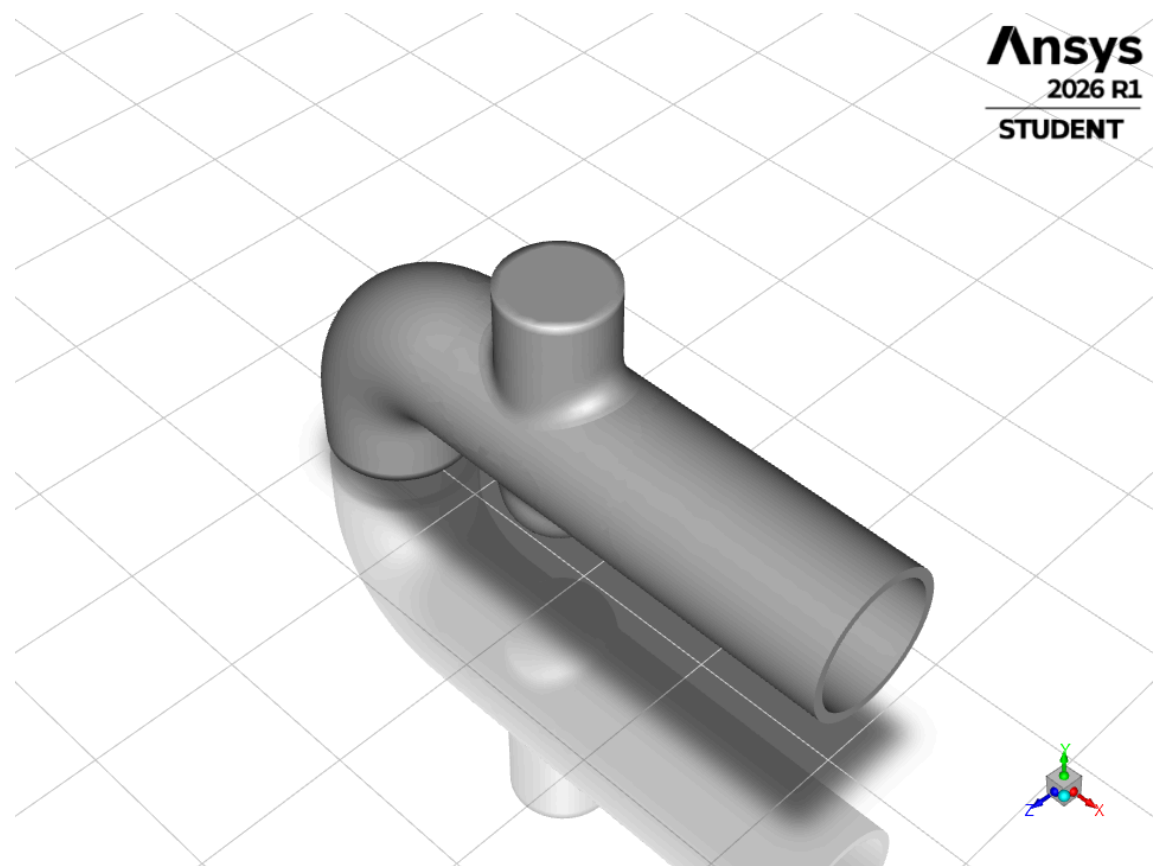
Plots

Residuals



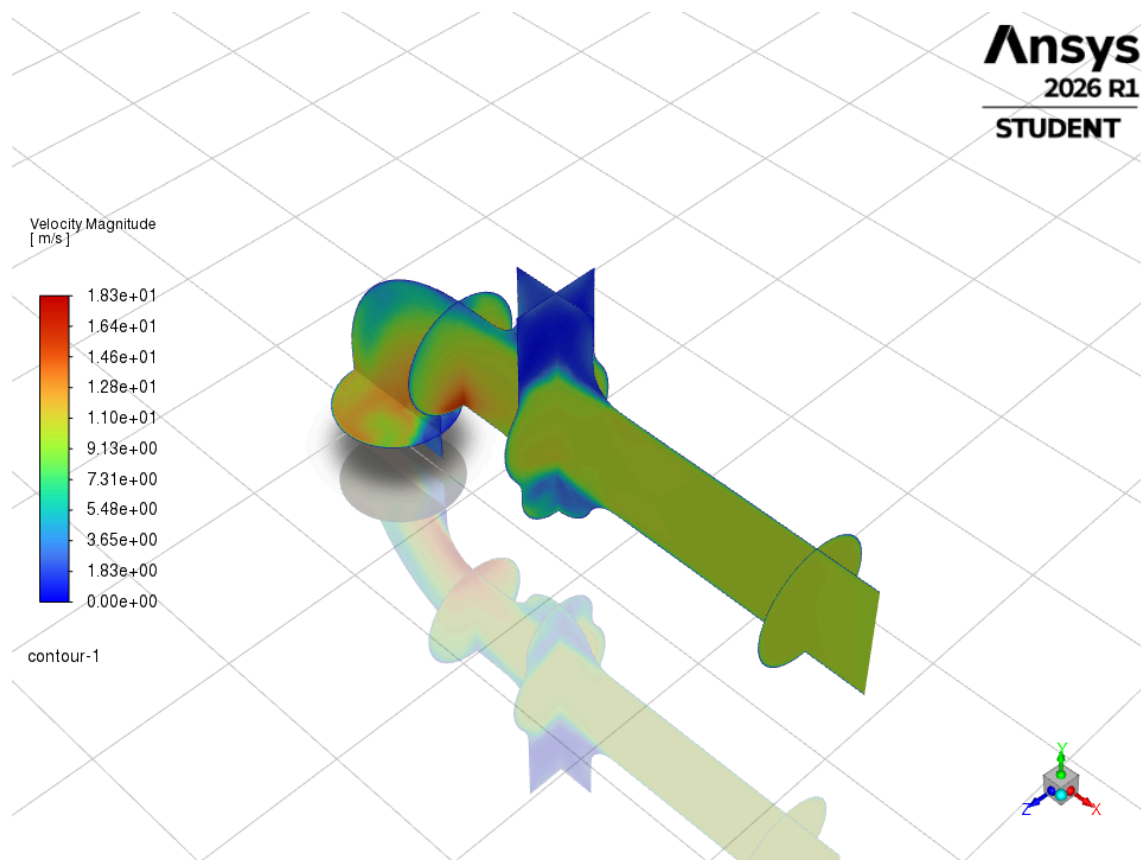
Mesh

mesh-1



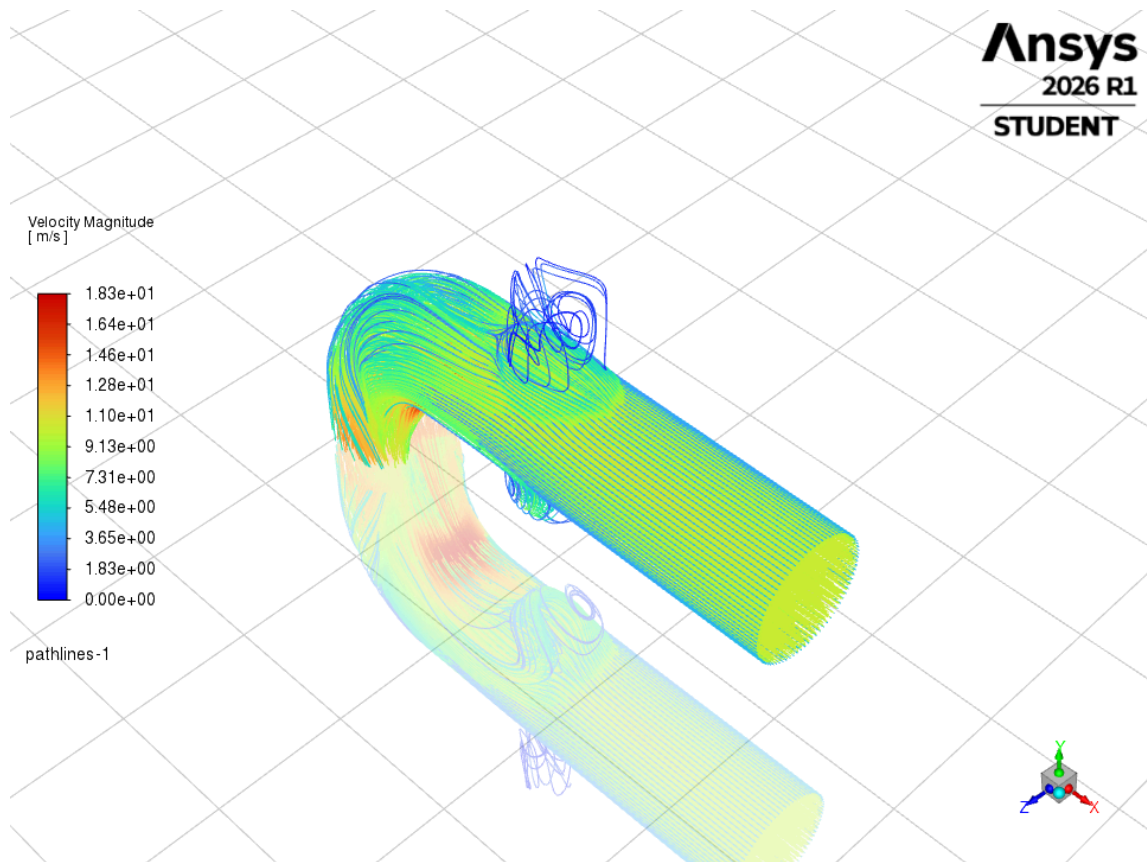
Contours

contour-1



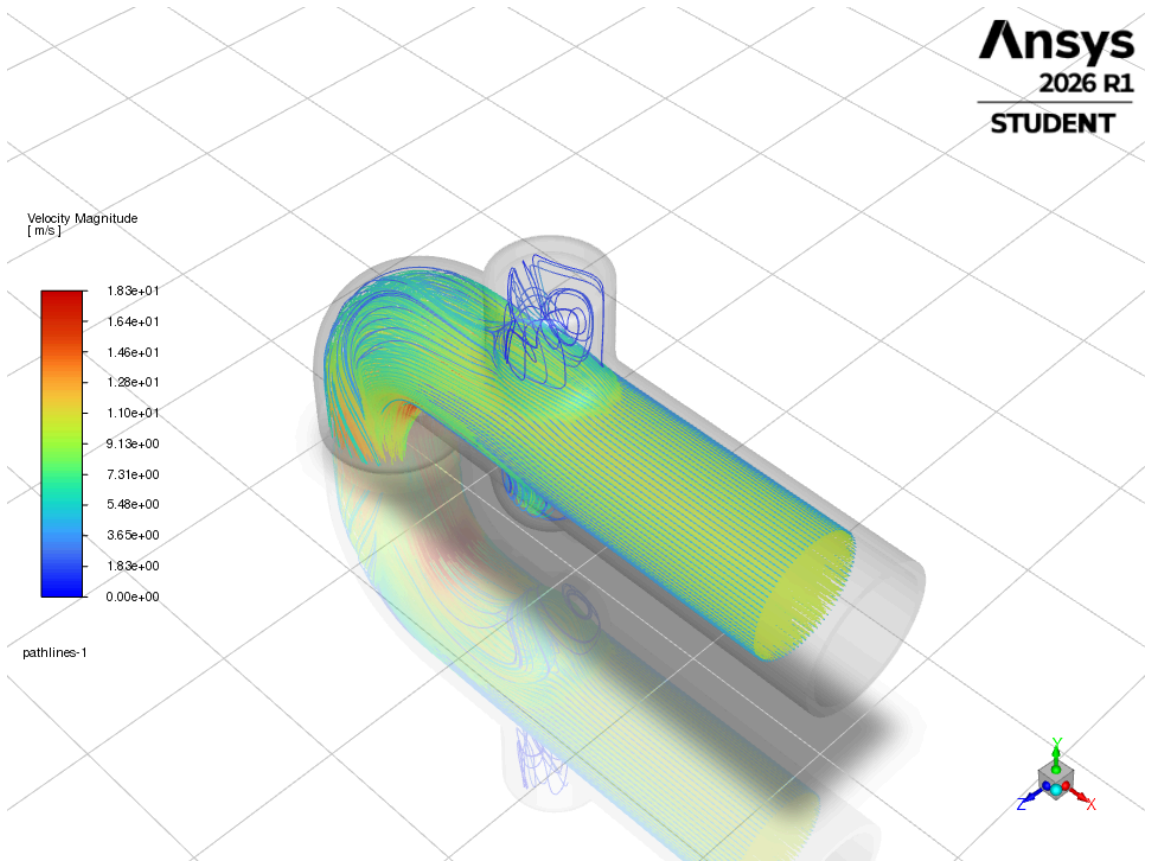
Pathlines

pathlines-1



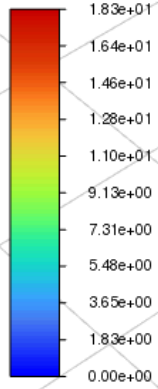
Scenes

scene-1



scene-1

Velocity Magnitude
[m/s]



pathlines-1

